Experiment 18 :Experiment to understand KNN algorithm for a given data set

Description:

The program demonstrates the K-Nearest Neighbors (KNN) classification algorithm using a given dataset. The data is divided into training and testing sets, and the KNN model is used to classify the test data.

Program:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
# New dataset
data = {
    "Height": [5, 6, 5, 7, 6, 8, 7, 9, 8, 10, 9, 11, 10, 12, 11],
    "Weight": [50, 55, 52, 65, 60, 72, 68, 80, 75, 88, 85, 95, 92, 105, 100],
    "Class": [
        "Small", "Small", "Small", "Medium", "Medium",
        "Medium", "Medium", "Large", "Large", "Large",
        "Large", "Large", "Large", "Large", "Large"
    ]
}
df = pd.DataFrame(data)
print("Dataset:")
print(df)
print("\nDataset Information:")
df.info()
print("\nClass Count:")
print(df["Class"].value_counts())
# Features and target
features = df[["Height", "Weight"]].values
label = df["Class"].values
# Split dataset
```

```
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)
# Create KNN model
model = KNeighborsClassifier(n_neighbors=3)
# Train model
model.fit(x_train, y_train)
# Prediction
y_pred = model.predict(x_test)
# Accuracy
print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))
# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

OUTPUT:

```
Dataset:
    Height  Weight   Class
0        5      50   Small
1        6      55   Small
2        5      52   Small
3        7      65  Medium
4        6      60  Medium
5        8      72  Medium
6        7      68  Medium
7        9      80   Large
8        8      75   Large
9       10      88   Large
10       9      85   Large
11      11      95   Large
12      10      92   Large
13      12     105   Large
14      11     100   Large

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Height  15 non-null     int64
 1   Weight  15 non-null     int64
 2   Class   15 non-null     str
dtypes: int64(2), str(1)
memory usage: 372.0 bytes

Class Count:
Class
Large     8
Medium    4
Small     3
Name: count, dtype: int64

Accuracy:
1.0

Confusion Matrix:
[[2 0]
 [0 1]]

Classification Report:
              precision    recall  f1-score   support

       Large       1.00      1.00      1.00         2
       Small       1.00      1.00      1.00         1

    accuracy                           1.00         3
   macro avg       1.00      1.00      1.00         3
weighted avg       1.00      1.00      1.00         3
```